

1.1 Statistics, Data, and the Role of Probability

Statistics and Data

Statistics is the science of collecting, organizing, presenting, analyzing, and interpreting data for the purpose of drawing meaningful conclusions and supporting informed decision-making. Every statistical investigation begins with a **data set**, which is a collection of observations, measurements, or numerical values obtained from a population or a sample.

A data set may consist of the heights of students in a class, daily temperatures recorded in a city, annual household incomes, the number of defective products manufactured in a factory, or examination scores of students. Regardless of its origin, the primary objective of statistics is to extract useful information from the data and to describe its essential characteristics in a concise and systematic manner.

Since a data set often contains a large number of observations, examining each observation individually is neither convenient nor efficient. Instead, statistics summarizes the entire data set by means of certain numerical quantities known as **statistical characteristics** or **statistical measures**. These measures provide compact descriptions of different features of the data and facilitate comparison, interpretation, and decision-making.

Among the most commonly used statistical characteristics are measures of central tendency (such as the mean, median, and mode), measures of dispersion (such as the variance and standard deviation), moments, coefficients of skewness and kurtosis, correlation coefficients, regression coefficients, and many others. Each of these measures describes a particular aspect of the data, such as its central location, variability, shape, or the relationship between two or more variables.

Although a statistical characteristic is expressed as a single numerical value, it is computed using all the observations in the data set. Thus, no single observation determines its value; rather, every observation contributes to its computation. A statistical characteristic therefore serves as a concise numerical summary that represents a particular feature of the entire data set.

Illustrative Example

Consider the following data representing the marks obtained by five students in a mathematics examination:

50,50,50,60,60.

The sample mean is

$$\bar{x} = \frac{(50+50+50+60+60)}{5} = \frac{(50 \times 3 + 60 \times 2)}{5} = 54.$$

Notice that the mark 50 occurs three times, whereas the mark 60 occurs twice. Instead of treating each observation separately, identical observations may be grouped according to the number of times they occur. This number is called the **frequency** of the observation.

The above computation shows that the sample mean can be viewed as a weighted average, where the weights are the relative frequencies of the observed values:

$$\bar{x} = \frac{(50 \times 3 + 60 \times 2)}{5} = 50 \times \frac{3}{5} + 60 \times \frac{2}{5} = 54.$$

This interpretation naturally leads to the concept of probability.

From Data to Probability

In descriptive statistics, statistical characteristics are computed directly from observed data. In many practical situations, however, the observed data are regarded as realizations of a random experiment or a random phenomenon. In such cases, the relative frequencies observed in a sample are interpreted as estimates of the probabilities of the possible outcomes/ events.

For the examination scores considered above, there are only two distinct values:

Mark	Relative Frequency
50	(3/5)
60	(2/5)

If these relative frequencies are interpreted as probabilities, the sample mean becomes

$$E(X) = \sum_{i=1}^2 x_i P(x_i) = x_1 P(x_1) + x_2 P(x_2) = 50 \times \frac{3}{5} + 60 \times \frac{2}{5} = 54.$$

which is precisely the **expected value** of the random variable X .

More generally, many statistical characteristics can be interpreted as weighted averages: in descriptive statistics the weights are relative frequencies, whereas in probability theory they are probabilities.

1.2 Representing Data with Random Variables

Statistics is primarily concerned with the collection, analysis, interpretation, and presentation of data, while probability provides the mathematical framework for describing the randomness and uncertainty inherent in data. Together, observed data and their associated probabilities enable statisticians to perform calculations, make estimates, and draw meaningful inferences about a phenomenon or population. Thus, probability forms the theoretical foundation upon which statistical methods and inference are built.

To apply the mathematical tools of probability in statistical analysis, it is convenient to represent the data as numerical values corresponding to the events generated by a random experiment using a random variable. Since every random experiment is associated with a sample space, and events are defined as subsets of that sample space, a variable whose values describe the events of a random experiment is called a random variable. Because the values assumed by a random variable describe the events of a random experiment, they are inherently uncertain, and this uncertainty is quantified through probabilities. Thus, a random variable provides a numerical representation of the events of a random experiment, while the observed values (or realizations) of the random variable constitute the data used in statistical analysis.

It should be noted, however, that not all possible events in a random experiment can necessarily be represented directly by a random variable. Moreover, to develop probability theory in a mathematically rigorous manner, it is necessary to specify a collection of events on which probabilities are defined. This leads to the concepts of **Boolean algebra** (or a field of sets) and, more generally, a **σ -algebra (sigma-algebra)**, which provide the mathematical structures required for defining probability measures and establishing the foundations of modern probability theory and statistics.

1.3 Mathematical Structures for Probability and Random Variables

Boolean Algebra (or just Algebra) and σ -Algebra (Sigma-Algebra)

A Boolean algebra, say $\mathcal{A}$, on a sample space Ω is a collection of subsets (called events) of Ω that satisfies three properties:

- i. The sample space Ω itself is in the collection, i.e, $\Omega \in \mathcal{A}$.
- ii. The collection is closed under complementation: if A is in the collection, then A^c is also in the collection, i.e, if $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$.
- iii. The collection is closed under union, i.e, if $A_1 \in \mathcal{A}$ and $A_2 \in \mathcal{A}$, then $A_1 \cup A_2 \in \mathcal{A}$.

A sigma-algebra differs from an algebra in that the above (iii) property is extended to countable unions, i.e., if $A_1, A_2, A_3, \dots, A_i \dots \in \mathcal{A}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$.

Probability Function

A probability function $P[\cdot]$ is a set function with domain $\mathcal{A}$ (an algebra of events or sigma-algebra) and counterdomain the interval $[0, 1]$ which satisfies the following axioms:

- i. Non-negativity: For every event $A \in \mathcal{A}$,
 $P[A] \geq 0$.
- ii. Normalization: The probability of the sure event (the sample space Ω) is 1, i.e.,
 $P[\Omega] = 1$.
- iii. Countable additivity (for mutually exclusive events): For any countable sequence of mutually exclusive events $A_1, A_2, A_3, \dots, A_i \dots \in \mathcal{A}$ such that $A_i \cap A_j = \emptyset$ for $i \neq j$ and if the union $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$, then

$$P\left[\bigcup_{i=1}^{\infty} A_i\right] = \sum_{i=1}^{\infty} P[A_i].$$

Note that the axioms are reasonable assumptions generally accepted as true. Irrespective of whether probabilities are interpreted a priori, posteriori, or subjectively, these axioms should be universally agreed upon; otherwise, the concept of probability would lose its mathematical consistency and reliability.

The above definition of probability is a mathematical definition; it tells us which set functions may be considered probability functions. It does not tell us what value the probability function $P[\cdot]$ assigns to any specific event A . To assign actual numerical probabilities, one must introduce additional assumptions or construct an appropriate model of the random experiment being studied, for example, by specifying physical symmetries, appealing to empirical relative frequencies, or adopting an underlying theoretical distribution. Only after such modeling can the probability function be fully determined for the situation at hand.

Limitations of Algebras

An algebra (or σ -algebra) of events $\mathcal{A}$ cannot generally contain all subsets of a sample space Ω because:

- a. Non-measurable sets exist when the sample space is uncountable.
- b. Assigning probabilities to all subsets leads to contradictions with additivity.
- c. The collection of *all* subsets is too large and contains “pathological” sets (that break the axioms of probability or measure theory).
- d. Probability theory requires sets to be measurable, and not all subsets are measurable.

Thus, an algebra must be a carefully chosen collection of subsets (events) that ensures the probability function is well-defined, consistent with axioms, and usable.

Probability Space

A probability space is a mathematical framework used to model random experiments. It is defined as a triple $(\Omega, \mathcal{A}, P[\cdot])$, where:

- i. Ω is the sample space, the set of all possible outcomes of the experiment.
- ii. $\mathcal{A}$ is a σ -algebra (sigma-algebra) of subsets of Ω , representing all measurable events.
- iii. $P[\cdot]$ is a probability function that assigns a probability $P[A]$ to each event $A \in \mathcal{A}$, satisfying the axioms of probability (non-negativity, normalization, and countable additivity).

The probability space serves as the foundation for probability theory. It allows for precise calculation of probabilities, modeling of random processes, and the development of further concepts such as conditional probability, independence, and random variables.

Random Variable

A random variable, denoted by $X(\cdot)$ or simply X , for a given probability space $(\Omega, \mathcal{A}, P[\cdot])$, is a function with domain Ω and codomain the real line $\mathbb{R}$. The function $X(\cdot)$ must satisfy the condition that for every real number r , the set $A_r = \{\omega: X(\omega) \leq r\}$ belongs to $\mathcal{A}$, meaning it represents an event. In this way, a random variable provides a way to describe events numerically.

Essentially, a random variable allows us to translate the abstract events of an experiment into numeric quantities that can be analyzed statistically.

Example:

Suppose

$\Omega = \{\text{HHHH, HHHT, HHTH, HHTT, HTHH, HTHT, HTTH, HTTT, THHH, THHT, THTH, THTT, TTHH, TTHT, TTTH, TTTT}\}$ represents the set of outcomes from flipping a coin four times. A random variable X could assign to each outcome the number of heads in that sequence. In this case, instead of needing all $2^{16} = 65,536$ subsets of Ω to describe events, we only need 5 subsets corresponding to $X=0, 1, 2, 3, 4$ heads.

Thus, a random variable uses Ω and $\mathcal{A}$ to define events in terms of real numbers. Using the random variable, one can assign probabilities to events through functions such as the cumulative distribution function (CDF), probability mass function (PMF), or probability density function (PDF). This approach allows us to model a random experiment by making assumptions about the random variable and its distribution, often bypassing the need to explicitly describe the full probability space.

In essence, a **probability space** provides the foundational structure. A random variable maps outcomes and events from this space to real numbers, the **CDF** summarizes the distribution of these values, and the **PDF** (or **PMF**) quantifies probabilities for individual values or intervals. Together, they form the core tools for modeling and analyzing uncertainty.

1.3 Why Study Probability Distributions?

A random variable provides a numerical representation of the events of a random experiment. However, knowing only the possible values that a random variable can assume is not sufficient for statistical analysis. To fully describe the random variable, we must also specify how likely each possible value (or range of values) is. This information is provided by its **probability distribution**.

A **probability distribution** specifies all possible values of a random variable together with the probabilities associated with those values. It is the mathematical object that translates the probability function defined on the events of a probability space into probabilities associated with the values of a random variable. For discrete random variables, this is accomplished through the **probability mass function (PMF)**, while for continuous random variables it is described by the **probability density function (PDF)**. In both cases, the **cumulative distribution function (CDF)** provides a unified description of the distribution.

Once the probability distribution of a random variable is known, all of its important characteristics can be derived directly from it. These include measures of central tendency (such as the expected value), measures of dispersion (such as the variance and standard deviation), higher-order moments, coefficients of skewness and kurtosis, and many other quantities that summarize the behavior of the random variable.

This extends the ideas introduced in the previous sections. In descriptive statistics, statistical characteristics are computed directly from observed data, with relative frequencies serving as

weights. In probability theory, these relative frequencies are replaced by probabilities, and the corresponding statistical characteristics become theoretical quantities of the probability distribution. Thus, probability distributions provide ***the mathematical bridge*** between **observed data** and **the theoretical models** used to describe random phenomena.

Because statistical inference seeks to draw conclusions about populations and random processes from observed samples, it necessarily relies on probability distributions. Methods such as parameter estimation, confidence intervals, hypothesis testing, regression analysis, analysis of variance, stochastic process modeling, and many modern statistical and machine learning techniques all depend on probability distributions to describe random variation and quantify uncertainty.

For these reasons, probability distributions occupy a central position in probability theory and mathematical statistics. A thorough understanding of probability distributions is therefore indispensable for the study of modern statistics and its applications in science, engineering, economics, medicine, and the social sciences.